

Complex Analysis: M. Math I: Final Examination

April 30, 2025.

Maximum Marks 60

Maximum Time 3:00 hrs.

- State any result you use or assume. Brief and clearly expressed answers are appreciated.
- ∂S is the boundary of $S \subset \mathbb{C}$. $\mathbb{D}$ is the unit disc, $D(0, R)$ is a disc centered at 0 and of radius R . $\mathbb{H}$ is the upper half plane.
- Questions are divided in Group A and Group B.

Group A (Maximum marks 35. Total marks 39.)

- (1) Show that

$$z \mapsto -\frac{1}{2} \left(z + \frac{1}{z} \right)$$

is a bi-holomorphic function from $\mathbb{D}^+$ to $\mathbb{H}$, where $\mathbb{D}^+ = \{z \in \mathbb{C} \mid |z| < 1, \Im z > 0\}$. [8]

- (2) (a) Show that $\text{SL}(2, \mathbb{R})$ acts transitively on $\mathbb{H}$.

(b) Let $\{a_n\}$ and $\{b_n\}$ be two disjoint sequences in $\mathbb{C}$ having no limit points. Stating and using an appropriate theorem, show that there exists a meromorphic function f on $\mathbb{C}$ which vanishes exactly at $\{a_n\}$ and have poles exactly at $\{b_n\}$, counting multiplicities for both zeros and poles. If f_1 and f_2 are two such functions, what is the relation between f_1 and f_2 ? [4+5=9]

- (3) (a) Show that there is an automorphism of $\mathbb{D}$ which does not have any fixed point.

(b) Show that if an automorphism f of $\mathbb{D}$ has two fixed points then f is the identity map. [4+5=9]

- (4) State and prove a Jensen's formula for the following situation: Let f be a meromorphic function on an open set Ω containing $\overline{D(0, R)}$ for some $R > 0$. Suppose that f has a simple zero at $a \in D(0, R)$ and a simple pole at $b \in D(0, R)$ and f has no other zeros or poles in $\overline{D(0, R)}$. [Hint: Use ψ_α functions.] [13]

P.T.O.

Group B (Maximum marks 25. Total Marks 28.)

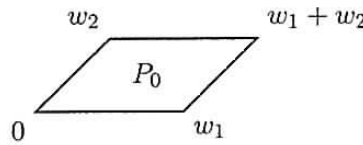
- (5) Let f be a nonconstant meromorphic function on $\mathbb{C}$ (which includes all entire functions).

Suppose that $f(z + w_1) = f(z)$ and $f(z + w_2) = f(z)$ for some $w_1, w_2 \in \mathbb{C}$ such that

$$\frac{w_1}{w_2} \notin \mathbb{R}$$

- (a) Show that f has infinitely many zeros in $\mathbb{C}$.

Let P_0 be the open parallelogram with vertices $0, w_1, w_1 + w_2, w_2$ as drawn below:



Suppose that $a \in P_0$ is a simple zero and $b \in P_0$ is a simple pole of f and f has no other poles or zeros in $\overline{P_0}$.

- (b) Show that

$$(*) \quad a - b = mw_1 + nw_2 \text{ for some } m, n \in \mathbb{Z}.$$

[Hint: $\int_{\partial P_0} z f'(z)/f(z)$ gives one side of $(*)$. Then compute this integral explicitly, consider the curve $f \circ \gamma_1$ etc., where γ_1 is the parametrization of a side.] [4+8=12]

- (6) Let F be an entire function and $F|_{\partial \mathbb{D}} = f$, a map on the unit circle. For $n \in \mathbb{Z}$, let

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{i\theta}) e^{-in\theta} d\theta.$$

Consider the following statements.

A. F maps $\mathbb{R}$ to $\mathbb{R}$ i.e. $F(\mathbb{R}) \subseteq \mathbb{R}$.

B. $\hat{f}(n) \in \mathbb{R}$ for all $n \in \mathbb{Z}$.

$$C. F(\bar{z}) = \overline{F(z)}.$$

Show that $A \iff B$ and $A \iff C$.

[4+4=8]

- (7) A holomorphic function $f : \mathbb{D} \rightarrow \mathbb{C}$ satisfies the following:

(a) f extends continuously to $\partial \mathbb{D}$,

(b) $|f(z)| = 1$ for $z \in \partial \mathbb{D}$,

(c) f has a simple zero at a , it has no other zeros on $\overline{\mathbb{D}}$ and $f(0) = a$.

Show that $f = \psi_a$ where $\psi_a(z) = (a - z)/(1 - \bar{a}z)$.

[8]